Automatic feedback for OPP texts

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Abstract

This study explores an automatic procedure that provides feedback to the Dutch Individual Development Perspective Plans (Ontwikkelingsperspectiefplannen; OPPs). State-of-the-art large language models (LLMs) are employed to evaluate OPP drafts while following a 17-criterion quality rubric, scoring the OPP drafts on each criterion and outputing a ranked list of revision suggestions. Our experiments show that DeepSeek V4 Flash produces more tailored quality feedback than Gemini 3.8 Flash, which could be associated with a higher "proficiency" in Dutch and longer reasoning process. We also analyse the actionability and tone of the LLM feedback, as well as reasons for its misalignment with human expectations. Based on the analysis, we identify potential solutions to problems observed in this study, including a multimodal approach.

1 Introduction

In this study, we explore an AI-assisted procedure to improve the Dutch Individual Development Perspective Plans (Ontwikkelingsperspectiefplannen; OPPs). These are formal documents created by education professionals when a school pupil needs more support than what is usually considered basic. This can include children with language learning difficulties, as well as social and emotional delays. A carefully prepared OPP document is important for schools to take the right actions in supporting the child.

Large language models (LLMs) such as ChatGPT, DeepSeek, and Mistral are pretrained on enourmous quantity of human data, and have shown surprising expertise in domains such as .... If LLMs could be effectively involved in preparing and improving OPP documents, it would greatly benefit early and primary education.

We are interested in the extent to which state-of-the-art LLMs can follow a quality rubric and provide feedback to OPP drafts. We ask the following questions: (i) To what extent do state-of-the-art LLMs capture the key quality issues of OPP drafts?[1] (ii) Is the feedback provided by the models useful for improving OPP texts? (iii) What can we do to improve the effectiveness of this AI-assisted procedure?

The first two questions deal with different components of the procedure. On one hand, the LLM should be capable enough to deliver meaningful quality assessment that aligns with human expectations. On the other hand, the feedback should also be actionable, providing concrete suggestions that guide the education professional through improving the OPP texts. We are interested in whether and how we can prompt LLMs to meet both requirements.

2 Experimental Design

We explored three output options:

1. Direct ranking, paired with natural language output instead of JSON.
2. Direct ranking, output as JSON.
3. Separate scores, output as JSON.

3 Experiments

We evaluated two state-of-the-art LLMs: Gemini 3.8 Flash (Google, 2026), which is one of the best LLMs for instruction following; and DeepSeek V4 Flash (DeepSeek-AI et al., 2026), one of the best LLMs for the Dutch language. We accessed these models through OpenRouter[2].

We tested the two LLMs on 24 OPP documents. The order of the documents was randomised. Since the OPP texts are in PDF files, we passed the files

[1] In a full-length study, I would compare model performance with a concrete ground truth set by education professionals.

[2] https://openrouter.ai/

through Cloudflare Workers AI (`cloudflare-ai`), a PDF parser provided by OpenRouter, before sending them to the test LLMs. While the Gemini model itself is capable of processing PDFs, we opted for an independent PDF parser to ensure that the two test models receive the exact same prompt. To assess the reliability of this pipeline, we prompted both models to briefly explain 5 randomly selected OPP files. Both models recounted most of the details in the provided documents, confirming that the PDF parser worked as expected and the test LLMs' understandng of the OPP texts would not be a factor that influence their quality assessment.

As for the (maximum) number of top-ranked weakpoints, we started with 5: This number of suggestions should provide a good starting point for revising the OPP sections without coming across as overwhelming.

For comparability, we used the same hyperparametres for the two models. Specifically, both temperature and top p were set to 1. These settings indicate that the models would be likely to provide the same answer when prompted again, and that they always output an answer with the highest probability. Reasoning was enabled for both models, leaving the models enough space to think through the task before outputing their ranked lists of criteria that the OPP texts met insufficiently.

To guide the LLMs to produce useful, actionable quality feedback, we designed a prompt that sets this goal explicitly from the start. The prompt also includes detailed steps with regard to how to produce the ranked list we ask for, as well as a specific JSON structure to format the ranked list. The full prompt can be found in the Appendix.

Results. Gemini 3.8 Flash ranked "Integratief beeld" as the number one priority for all but one OPP document, followed by "Ondersteuningsbehoeften" as the second criterion for more than half of the documents (16 out of 24). Its explanations for "Integratief beeld" also sound similar. Additionally, since these two criteria are the only two with the highest weight, one may doubt whether the model ranked them higher in most cases because such feedback is unlikely to be wrong.

On the other hand, while DeepSeek V4 Flash gave "Integratief beeld" top priority for more than half of the documents (15 out of 24), its overall assessments show much more variability than the Gemini model. We also found that the DeepSeek model produced much longer reasoning texts (more than 15K tokens vs under 10K tokens) despite the two models sharing the same hyperparamtres.

4 Discussion

Tailored feedback from DeepSeek. Our experiments indicate that DeepSeek V4 Flash is capable of providing more tailored feedback to the OPP texts than Gemini 3.8 Flash. This could be associated with (i) the DeepSeek model's superior performance in the Dutch language, and (ii) the much longer reaonsing texts the model produced. One of the challenges of the task is the many steps it involves: reading and interpreting the quality rubric; reading the OPP document; giving a score to each criterion; calculating priorities; using the given JSON template to format its output. For an LLM without prior training targeting this task, it is plausible that a longer reasoning process leads to more tailored quality feedback.

Actionability of LLM feedback. Both models identified "Integratief beeld" as an area that more than half of the OPP documents need improvement. While this may reflect a certain level of truth, it is questionable whether putting it on the top of the list is genuinely helpful for education professionals creating the OPP documents. A useful integrative view is something that heavily relies on the individual parts of a OPP document. In a real-life OPP revision workflow, the education professional may deal with it after polishing other OPP sections. An actionable plan would probably put it at the bottom instead of the top while still emphasising its importance.

In this study, we explicitly requested LLMs to rank quality criteria in terms of importance; our findings suggest that importance may not be interchangeable with "priority" in this context. Future studies could therefore investigate alternative prompt templates that lead to a actionable quality feedback.

The tone of LLM assessment. A closer look at the model outputs reveals that Gemini 3.8 Flash could sound harsh in its explanations of the scores. On one occassion, for example, the model declares, "The educational needs consist entirely of standardized, copy-pasted checklist phrases ...." The DeepSeek model, on the other hand, adopts a milder, more supportive tone: "The support needs are listed and somewhat concrete, but they remain

fairly generic ....” A harsh tone could appear unwelcome to the OPP writer, and subsequently discourage them from following the AI's suggestions. It is therefore worth considering explicitly instructing the LLM to return its assessment in an appropriate tone.

LLMs' understanding of OPP document. Despite sufficient understanding of the *contents* of the documents, the models show fudamental differences from humans with regard to how they perceive and treat the contents. From a human's perspective, it is clear that the documents follow the same structure, containing both set metrics and free-writing parts; and it is primarily the quality of the free writing that we are tasking LLMs with. However, since our test models receive the documents in markdown format, they assume that “OPP texts” in the instructions mean the entire document, including the parts that are supposedly unchangeable.

This finding indicates that vision-language models with integrated PDF processing capabilities may be more suitable for this task. To test this hypothesis, we reran the experiment with Gemini 3.8 Flash, using its native PDF processing engine and 5 randomly selected OPP documents. The model produced similar quality feedback in this settings as compared to the previous setting that uses an external PDF parser. It placed “Integratief beeld” at the top of the list for all 5 OPP documents, and did not distinguish document parts that are standardised and those that consist of free writing. A possible reason is that both the Gemini model and the external parser convert PDF documents to plaintext, where the differences between the sections are obscured. It could also be becuase current LLMs have not received enough training regarding the structure of OPP documents.

There are several directions future research could consider with regard to improving LLMs' understanding of OPP document structures and the quality assessment task. First of all, the quality rubric could be more elaborate, including explanation and examples for each criterion. This adjustment would leave less room for the LLM to interpret the criteria on its own, and thus potentially resulting in feedback that targets the right parts of an OPP document. Second, future studies could investigate different ways to present the OPP document. Both textual and multimodal approaches are worth considering. For a multimodal approach, one could use images to present the standardised parts, and texts the free writing parts.

Limitations

In the experiments, we provided the LLMs the full OPP documents so that they would have full context for quality assessment. However, as addressed in the Discussion section, there are alternative ways to present the documents apart from passing them directly through a PDF parser. A plausible future direction is to investigate the best ways to structure OPP texts for LLM-assisted quality assessment. One may also be interested in whether a more concise OPP input would result in better feedback from the LLM.

We also acknowledge that the current quality rubric may not be sufficiently self-explanatory for LLMs, and we did not optimise hyperparametres such as temperature and top p for the task. While the results of the experiments provide useful insights for improving the AI-assisted OPP revision pipeline, they do not necessarily reflect the best performance current LLMs could reach in our task.

Another limitation of the quality rubric is that a total score of 56 if unintuitive. For a more human-centred approach, it could be helpful mapping the full score onto a 0–10 or 0–10 scale.

References

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